Q11: Why Models Hallucinate, and Whether They Can Know They Are Unsure

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Learning objectives

- Explain the statistical, data-driven, and training-objective mechanisms that cause large language models (LLMs) to produce confident but false statements ("hallucinations").
- Distinguish *epistemic* from *aleatoric* uncertainty and connect each to concrete failure modes and measurement techniques (calibration, selective prediction, semantic entropy).
- Critically assess whether and how a model can *know* it is uncertain, and what interventions (abstention, retrieval, verbalized confidence, fine-tuning) help.

1 What we mean by "hallucination"

In the LLM literature, a hallucination is text that is fluent and syntactically well-formed but factually unsupported—either inconsistent with the provided source (*faithfulness* or intrinsic hallucination) or inconsistent with world knowledge (*factuality* or extrinsic hallucination). The survey by Ji et al. [3] formalizes this distinction across natural language generation tasks, and Maynez et al. [12] documented pervasive unfaithfulness even in abstractive summarization systems trained on curated data. The crucial feature for this module is that hallucinations are not random noise: they are the generically correct output of a system optimized to produce *plausible continuations*, not *true* ones. Understanding why they arise therefore requires examining the objective, the data, and the decoding procedure together.

2 Mechanism 1: next-token prediction rewards plausibility, not truth

An autoregressive LM is trained to maximize the likelihood of the next token given context, i.e. to approximate the conditional distribution $p(x_t \mid x_{<t})$ of human text. Nothing in this objective references the external truth of a proposition; it references how probable a continuation is *given the training corpus*. When a prompt enters a region of input space where the true answer is sparsely or inconsistently represented, the model still emits the most probable-*looking* completion, which may be a smooth interpolation of training fragments rather than a fact. This is the core intuition behind the recent OpenAI analysis by Kalai et al. [5], who argue that hallucinations are an expected consequence of the pretraining objective and are further *reinforced* by evaluation benchmarks that reward guessing over abstaining. Because standard benchmarks score a blank or "I don't know"

answer the same as a wrong answer (zero credit), the expected-score-maximizing strategy is to always produce a confident guess. The model learns the policy that its grader rewards.

A second structural point: language models are density estimators over sequences, and the calibration results of Guo et al. [2] for deep classifiers generalize—the alignment between a model's token probabilities and empirical correctness is not guaranteed and tends to degrade after certain fine-tuning steps. A confidently sampled token sequence is not an epistemic warrant.

3 Mechanism 2: data pressures—long tails, staleness, and false premises

Pretraining corpora are heavy-tailed: a vast number of entities, dates, and relations appear only a handful of times. For such long-tail facts the model has insufficient signal to memorize the correct association, so it reconstructs a statistically typical but possibly wrong value (e.g. plausible-sounding citations, invented DOIs). Corpora are also *stale*: knowledge acquired at a training cutoff cannot track the evolving world, so a model will confidently assert outdated facts. Finally, prompts frequently contain *false presuppositions*. The TruthfulQA benchmark of Lin et al. [10] shows that models often reproduce popular human misconceptions precisely because imitating the training distribution means imitating its errors; larger models can be *less* truthful on these adversarially selected questions, a phenomenon the authors call "inverse scaling" for truthfulness. Data, then, sets an upper bound: a model cannot reliably report a fact that is absent, contradictory, or systematically misrepresented in what it saw.

4 Mechanism 3: two kinds of uncertainty

To ask whether a model can "know" it is unsure we must separate two sources of uncertainty, a distinction imported from Bayesian modeling by Kendall and Gal [6].

- **Aleatoric uncertainty** is irreducible randomness in the data-generating process—e.g. genuinely ambiguous questions or open-ended continuations with many valid answers. More data does not remove it.
- **Epistemic uncertainty** reflects the model's lack of knowledge: regions where the parameters are underdetermined because training data were sparse. This is the uncertainty we *want* a model to recognize before hallucinating, and in principle it is reducible with more data or retrieval.

Hallucinations are most damaging when they occur under high epistemic uncertainty that the model fails to flag. A well-behaved system should raise a signal in exactly those regions. The practical difficulty is that a single softmax over tokens conflates both uncertainties and, worse, is computed *token-wise* rather than over *meanings*.

5 Mechanism 4: can the model tell that it is unsure?

There is encouraging evidence that the information needed to detect uncertainty is partially present inside the model.

Internal confidence and calibration. Kadavath et al. [4] show that LLMs can be prompted to estimate the probability that their own answer is correct (a "P(True)" head) and that this self-evaluation is reasonably calibrated on many tasks, improving with scale. This suggests the model carries a usable, if imperfect, internal signal of correctness distinct from the raw generation probability.

Semantic, not lexical, entropy. Naively, we might measure uncertainty as token-level entropy, but the same fact can be phrased in many ways, so lexical diversity overstates uncertainty. Kuhn et al. [7] introduce *semantic entropy*: sample several answers, cluster them by meaning using entailment, and compute entropy over meaning-clusters. High semantic entropy—many *mutually contradictory* answers—is a strong predictor of hallucination. Farquhar et al. [1] extend this to detect "confabulations" (arbitrary, incorrect generations) across free-form tasks, demonstrating that an unsupervised uncertainty estimate can flag likely fabrications without ground-truth labels.

Consistency as a proxy. A complementary black-box method, SelfCheckGPT [11], samples multiple responses and checks whether facts recur consistently; isolated, non-reproducible claims are treated as probable hallucinations. The underlying assumption is that genuinely known facts are stable under resampling, whereas confabulated ones vary.

Verbalized confidence. Models can also be trained or prompted to state confidence in words ("I am about 60% sure"). Lin et al. [9] find that fine-tuned models can emit verbalized probabilities that are moderately calibrated, offering a human-readable uncertainty channel. However, verbalized confidence can be gamed by stylistic cues and should be validated against outcome calibration.

Taken together, these results imply a qualified "yes": the model possesses latent signals correlated with its own unreliability, recoverable through self-querying, resampling over meanings, or lightweight probes. What it lacks is a *built-in incentive* to surface them, because the training and evaluation pipeline rewards fluent commitment.

6 Mechanism 5: interventions and their limits

Abstention / selective prediction. If an uncertainty score is available, the system can refuse to answer below a threshold, trading coverage for accuracy. This is classical selective prediction; the open problem is choosing thresholds that respect task-specific costs and avoid over-refusal.

Grounding via retrieval. Retrieval-augmented generation [8] reduces extrinsic hallucination by conditioning the output on retrieved documents, effectively converting an epistemic gap into a reading task. It does not eliminate hallucination: the model may ignore, misread, or over-generalize the evidence, and retrieval errors propagate.

Alignment with explicit abstention incentives. Because Kalai et al. [5] trace part of the problem to graders that penalize honesty, a direct fix is to redesign evaluation and reward so that a calibrated "I don't know" earns more than a confident error. Reinforcement learning from human feedback [13] can in principle instill such behavior, but reward models can themselves mis-specify honesty and induce sycophancy.

Calibration repair. Post-hoc methods (temperature scaling, following Guo et al. [2]) and semantic-entropy gating act at inference time and are cheap, but they correct the symptom rather than the data-level cause.

7 Summary and study prompts

Hallucination is the predictable output of maximizing textual plausibility under a heavy-tailed, stale, error-containing corpus, amplified by benchmarks that reward guessing. A model does carry internal signals of its own epistemic uncertainty—recoverable via P(True) self-evaluation, semantic entropy, sampling consistency, and verbalized confidence—but current objectives do not reliably compel it to act on them. Reducing hallucination therefore requires both *estimation* (detect uncertainty, distinguishing epistemic from aleatoric) and *incentive design* (reward honest abstention and grounding). Consider: (i) Why does token-level entropy overstate uncertainty, and how does semantic clustering fix it? (ii) Design a grading rule for a QA benchmark that makes abstention the optimal policy under uncertainty. (iii) When does retrieval *fail* to prevent hallucination despite correct evidence?

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